



MS-40M Pyranometer

Technical Specifications

ISO 9060:2018 Class C (Second class)

Sub-category "Spectrally flat"

Industrial sensor outputs (MODBUS RTU)

ISO 17025 certified calibration

Optional ventilator MV-01

The MS-40A is an ISO 9060:2018 Class C (Second Class) pyranometer which is based on EKO's universal sensor platform with Modbus 485 RTU output. This sensor can be used in many meteorological networks and professional small scale PV sites where solar radiation is taken seriously. The MS-40/40A/40M can be used with the optional MV-01 ventilator / heater or can be combined with the optional mounting kit for albedo measurements.

The MS-40M pyranometers are manufactured in a consistent way followed by strict quality inspection and performance evaluation. EKO provides a unique calibration compliant to the international standards defined by ISO/IEC17025/9847.

The sensor has a 5 years warranty with a 2 years re-calibration interval recommended and it is no longer necessary to change the desiccant.

	MS-40M
ISO 9060:2018	Class C
ISO 9060:1990	Second Class
Sub-category "Spectrally flat"	Compliant
Sub-category "Fast response"	Not compliant
Output	Digital (Modbus RTU)
Response time 95%	< 18 Sec.
Zero off-set a) 200W/m ²	+/- 12 W/m ²
Zero off-set b) 5K/hr	+/- 5 W/m ²
Complete zero off-set c)	+/- 17 W/m ²
Non-stability change/1 year	+/- 1.5 %
Non-linearity at 1000W/m ²	+/- 1 %
Directional response at 1000W/m ²	+/- 20 W/m ²
Spectral error	+/- 0.16 %
Temperature response -10°C to 40°C	+/- 3 %
Temperature response -20°C to 50°C	+/- 3 %
Tilt response at 1000W/m ²	+/- 1 %
Sensitivity	-
Impedance	-
Operating temperature range	-40 - 80 °C
Irradiance range	0 - 2000 W/m ²
Wavelength range	285 - 3000 nm
Ingress protection IP	67
Cable length	10 m

Power supply	12 - 24 VDC
Power consumption	0.2 - 0.3 W
Additional signal processing errors	+/- 1.5 W/m ²

Options	MS-40M
Cable length	20 / 30 / 50 m
Ventilation unit	MV-01
Albedo mounting kit	MS-albedo Kit

Specifications are subject to change without further notice.

EKO, MS-40M Pyranometer

Port Name: COM22 Refresh Time: 1000 ms
 Baudrate: 9600 Node Index: ID 81
 Parity: None Refresh Serial Ports ☐ Configuration Reading Success...

Node

	Address	Value		Address	Value
Serial Num.	0	S18118081	TC-c	30	0.000e+000
Sensor Model	8	MS-40M	TC-d	32	0.000e+000
Irrad. Min.	13	0 W/m ²	NL-k1	34	0.000e+000
Irrad. Max	14	1600 W/m ²	NL-k2	36	1.000e+000
Acqui. Mode	15	1	NL-k3	38	0.000e+000
Sensitivity	16	12.410 μ V/W/m ²	NL-k4	40	0.000e+000
Temp. Format	18	1	mb ID	42	81
Millivolts	19	-0.002 mV	mb Baud.	43	9600
Irradiance	21	-0.191 W/m ²	mb Par.	45	None
Temperature	23	-40.00 $^{\circ}$ C			
Password	25	*			
TC-a	26	1.000e+000			
TC-b	28	0.000e+000			

Read this Holding register

Stop Bit 2

Send Config. Log Data

(Temp. & Humidity Sensor)

Modbus Poll - Mbpoll1

File Edit Connection Setup Functions Display View Window Help

05 06 15 16 17 22 23 TC

Mbpoll1 Data Bits: 8

Tx = 244: Err = 0 ID = 82 F = 03: SR = 1000ms Parity: None

Stop Bits: 2
Baudrate: 9600

	Alias	00020	Alias	00030
0				
1				
2				
3				
4				
5				
6	Temp.	19.4783		
7		--		
8		66.8397		
9		--		